

AI Engineering Bootcamp — Industry-Level Program

Built on real job descriptions. Taught like a senior engineer would.

Prerequisites

- No prior coding experience needed
- A laptop with internet connection
- 2–4 hours daily commitment
- Hunger to get hired — nothing else required

Section 1 — Python for Engineers

- Python fundamentals engineers actually use on the job
- Data types, loops, functions, list comprehensions
- OOP basics — classes, objects, inheritance
- File handling — reading/writing JSON, CSV, TXT
- Virtual environments & dependency management (pip, venv, UV, Anaconda)
- Writing clean, readable, production-style code
- Exception handling & defensive programming

Project: python student management Tool

Section 2 — Backend Engineering & REST APIs

- Backend architecture — how real systems are built
- REST API design principles & best practices
- FastAPI — routing, middleware, dependency injection
- Request validation with Pydantic v2
- Authentication basics — API keys & JWT tokens
- Rate limiting & request throttling
- API versioning & documentation (Swagger/OpenAPI)
- Git branching strategy — feature branches, PRs, code review

Project: Production-ready REST API with auth & docs

Section 3 — Databases

- Relational database design & normalization
- Advanced SQL — joins, indexes, aggregations
- SQLAlchemy ORM — models, relationships, migrations
- Data modeling for AI applications

Project: Scalable User & Data Management System

Section 4 — LLM Engineering & AI Integration

- LLM internals — transformers, attention, tokenization
- OpenAI & Claude — API mastery
- Hugging Face Hub — loading & using open source models
- Running LLaMA 3, Mistral, Phi-3 locally with Ollama
- Advanced prompt engineering — system prompts, few-shot, CoT, ReAct
- Structured outputs — JSON mode, function schemas
- Token optimization & cost control
- Streaming responses & async LLM calls

- Multi-model routing — when to use which model

Project: Production AI Assistant with multi-model support

Section 5 — Embeddings, Vector Search & Semantic Systems

- Embedding models — how they encode meaning
- OpenAI text-embedding-3 (small & large)
- Hugging Face Sentence Transformers (open source & free)
- Cosine similarity, dot product, Euclidean distance
- Approximate nearest neighbor search (ANN)
- FAISS — local vector index, GPU acceleration
- Pinecone, ChromaDB— production vector databases
- Metadata filtering & hybrid search
- Chunking strategies — fixed, semantic, recursive

Project: Semantic Search Engine over real documents

Section 6 — RAG Engineering (Most In-Demand Skill)

- RAG architecture — ingestion, retrieval, augmentation, generation
- Advanced document loaders — PDF, DOCX, web scraping, databases
- Chunking & preprocessing pipelines
- Embedding generation at scale — batch processing
- Retrieval strategies — dense, sparse, hybrid
- Re-ranking with cross-encoders (Cohere, open source)
- Query expansion & HyperRAG techniques
- Context window management & stuffing strategies
- Prompt injection prevention & output guardrails
- Hallucination detection & mitigation
- RAG evaluation — RAGAS (faithfulness, relevancy, recall)
- Cost-optimized RAG — full open source stack (LLaMA + ChromaDB)

Project: Enterprise RAG Chatbot over company knowledge base

Section 7 — AI Agents & Agentic Workflows

- Agent architecture — perception, planning, action, memory
- ReAct, Plan-and-Execute, Reflexion patterns
- OpenAI function calling & tool use
- Building custom tools for agents (APIs, code execution, search)
- LangGraph — stateful, multi-step agent workflows
- Multi-agent systems with AutoGen & CrewAI
- Agent memory — short-term, long-term, episodic
- Human-in-the-loop agent design
- Async task queues with Celery & Redis
- Agent reliability — error recovery, retries, fallbacks
- Cost governance in agentic systems

Project: Autonomous AI Agent for business workflow automation

Section 8 — Fine-Tuning & Model Customization

- When to fine-tune vs prompt engineer vs RAG
- Dataset preparation & formatting (instruction tuning format)
- LoRA & QLoRA — parameter efficient fine-tuning
- Fine-tuning LLaMA 3 / Mistral on custom data
- PEFT with Hugging Face — hands-on training
- Experiment tracking with Weights & Biases
- Model evaluation — BLEU, ROUGE, human eval
- Merging & exporting fine-tuned models
- Serving fine-tuned models with Ollama / vLLM

Project: Domain-specific fine-tuned model (legal / medical / finance)

Section 9 — Deployment, MLOps & LLMOps

- Docker — containerizing AI applications
- CI/CD pipelines with GitHub Actions
- Deploying on Azure/AWS/GCP, Hugging Face Spaces
- Serving LLMs at scale with vLLM & TGI
- LLM observability — tracing, logging, evaluation with Langfuse
- Monitoring
- A/B testing AI models in production
- Cost tracking & budget alerts at scale
- Load testing AI endpoints

Project: Production-deployed AI system — fully monitored & scalable

Section 10 — AI System Design & Interview Preparation

- AI system design patterns used at top companies
- Designing RAG at scale — 1M+ documents, 10k+ users
- Vector database selection & tradeoffs at scale
- Latency optimization for real-time AI systems
- Caching strategies — semantic cache, prompt cache, response cache
- Designing multi-agent orchestration systems
- Real interview questions from Google, Microsoft, Anthropic, startups
- Portfolio & GitHub optimization for AI engineering roles
- Resume writing aligned to AI engineering job descriptions
- Mock coding interviews — Python + DSA essentials
- Mock AI system design interviews with full feedback
- Salary benchmarking & negotiation playbook

No project — 100% job-readiness focus

Section 11 — Final Capstone + Corporate Demo

Tools: Student chooses full stack based on their track

Choose your track:

AI SaaS Product — end-to-end AI-powered web product

- Enterprise Knowledge AI — internal company intelligence system
- AI Automation Platform — multi-agent business automation
- **Capstone requirements:**
- Production backend with auth & database
- LLM integration (OpenAI or fully open source)
- RAG pipeline with vector database
- Agentic workflows (at least one autonomous agent)
- Full cloud deployment with CI/CD
- LLM monitoring & observability
- Architecture document & cost analysis report
- Live demo to panel of industry engineers

Job Roles You Can Apply For After This Program

Role	What You'll Do
AI Engineer	Build LLM-powered products end to end
LLM Application Developer	Integrate LLMs into enterprise systems
RAG Engineer	Design and optimize retrieval-augmented systems
AI Agents Engineer	Build autonomous agentic workflows
GenAI Developer	Ship generative AI features across products
Backend + AI Developer	Strong backend with deep AI integration
MLOps / LLMOps Engineer	Deploy and monitor AI at scale

